

From Description Logics to (Virtual) Knowledge Graphs

João Pedro de Araújo Lima
Advisor: Renata Wassermann

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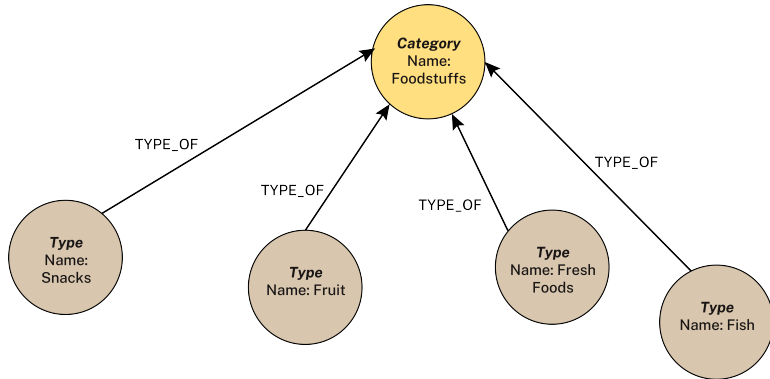
Section 1

Introduction and preliminary concepts

What is a Knowledge Graph?

Knowledge Graph

A knowledge graph (KG) can be broadly defined as a graph of data intended to accumulate and convey knowledge of the real world [Hogan et al., 2021].



What is a Virtual Knowledge Graph?

Virtual Knowledge Graph (VKG)

The Virtual Knowledge Graph (VKG) approach (also widely known in the literature as Ontology-Based Data Access (OBDA)) offers a powerful way to access and integrate data [Calvanese et al., 2016].

VKG: Architecture and Components

A typical VKG system relies on four core components working together [Calvanese et al., 2016]:

Ontology

Conceptual map of the domain: defines classes and properties using OWL or RDFS.

Data Sources

Underlying physical storage, most commonly relational databases accessed via standard drivers.

Mappings

Declarative rules in R2RML (W3C standard) linking data sources to ontology terms.

Queries

User requests in SPARQL, using ontology terms instead of rigid database schemas.

VKG: A Virtual RDF Graph

Ontology + Mappings = Virtual RDF Graph

Together, the ontology and the mappings define an RDF graph that does not need to be physically stored [Calvanese et al., 2016].

Materialization

- ▶ Generates and stores all triples in a triplestore
- ▶ Data duplication
- ▶ Update delays and maintenance costs

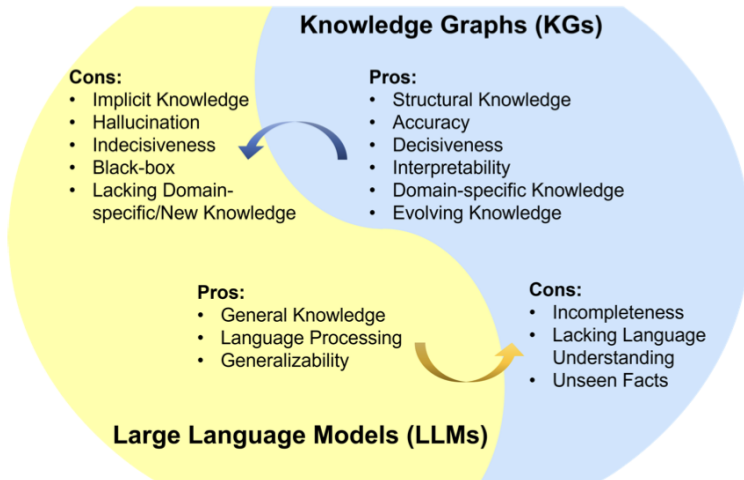
Virtual (e.g., Ontop)

- ▶ SPARQL translated to SQL on-the-fly
- ▶ No data duplication
- ▶ Leverages RDBMS maturity, security, and query optimization

Some Applications

- ▶ **Data integration:** Although Ontop connects to a single JDBC source, it can integrate multiple heterogeneous databases by relying on data federation systems such as Denodo, Dremio, Teiid, Apache Spark, or Trino;
- ▶ **Avoid data duplication:** By keeping data in its original sources, the VKG approach avoids unnecessary copies, which is particularly relevant under data privacy regulations such as the GDPR;
- ▶ **Facilities:** Domain experts could query data using domain terms, without relying on SQL experts;

Maybe another application...



Unifying Large Language Models and Knowledge Graphs: A Roadmap [Pan et al., 2024].

Language Models: LLMs

Large Language Models (LLMs)

Probabilistic models that predict the next token from context. Their outputs are predictions, **not verified facts**, and are prone to errors and hallucinations [Huang et al., 2025].

But Approaches such as OG-RAG [Sharma et al., 2025] show that **formal knowledge structures** reduce hallucinations in neural generation.

KG-enhanced LLMs

Goal

Use KGs to improve LLMs in downstream tasks [Pan et al., 2024].

Pre-training

KGs enhance the interpretability and knowledge expression of LLMs during training.

Inference

KGs provide up-to-date knowledge at inference time, removing the need to retrain the model.

Interpretability

KGs can serve as a mechanism for interpreting the model's reasoning.

LLM-augmented KGs

Motivation

Current KG methods struggle with incomplete graphs and with processing text corpora for KG construction [Pan et al., 2024].

LLMs can be used for

- ▶ **Embedding:** encoding textual descriptions of entities and relations
- ▶ **Completion:** text encoding or fact generation to complete KGs
- ▶ **Construction:** entity discovery and relation extraction
- ▶ **KG-to-text:** generating natural language descriptions of facts
- ▶ **Question answering:** bridging natural language questions and KG retrieval

Synergized LLMs + KGs

Goal

Mutual enhancement by unifying two inherently complementary techniques [Pan et al., 2024].

LLMs

- ▶ Linguistic flexibility
- ▶ Generalization

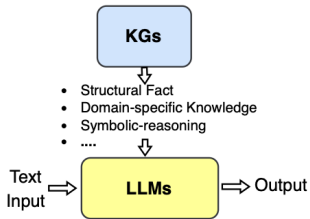
KGs

- ▶ Factual knowledge
- ▶ Explicit structure

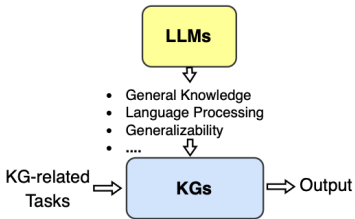
Result

A neuro-symbolic system combining statistical and symbolic reasoning [Pan et al., 2024, Yang et al., 2025].

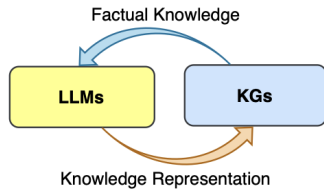
All approaches



a. KG-enhanced LLMs



b. LLM-augmented KGs



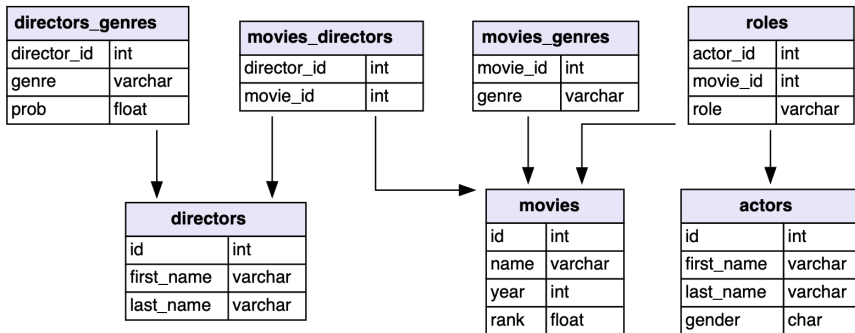
c. Synergized LLMs + KGs

Unifying Large Language Models and Knowledge Graphs: A Roadmap [Pan et al., 2024].

Section 2

Hands-on

IMDB Dataset



<https://relational.fel.cvut.cz/dataset/IMDb>.
<https://github.com/pedro0058/obda-imdb-tutorial>

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